

# KGGuard: Knowledge Graph-Enhanced Hallucination Detection for Large Language Models

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## Abstract

We present KGGuard (Knowledge Graph Guard), a novel model-agnostic hallucination detection system that addresses the critical challenge of semantic over-generation in Large Language Models (LLMs). Our approach innovates by bridging unstructured LLM outputs with structured knowledge representations through a multi-layered verification pipeline. KGGuard combines Named Entity Recognition (NER), relation extraction, and knowledge graph validation with confidence-weighted scoring and temporal validation. Unlike existing surface-level approaches, our system performs structural sanity checks and provides explainable AI capabilities by visualizing incorrect relationships in knowledge graphs. Experimental results demonstrate superior performance in detecting factual, contextual, and logical hallucinations while maintaining model-agnostic compatibility.

## 1 Introduction

The proliferation of Large Language Models has introduced unprecedented capabilities in natural language generation, yet it has simultaneously amplified the critical challenge of hallucination detection. Current hallucination detection methods primarily focus on surface-level probability analysis or simple fact-checking, failing to capture the deeper semantic and structural inconsistencies that characterize sophisticated hallucinations.

Our contribution addresses this gap through KGGuard, a system that translates unstructured LLM outputs into structured knowledge representations for comprehensive verification. This approach enables detection of not only factual inaccuracies but also logical inconsistencies and structural anomalies that traditional methods miss.

## 2 Related Work

Traditional hallucination detection approaches fall into several categories: (1) probability-based methods that analyze token confidence scores, (2) consistency-based approaches that compare multiple model outputs, and (3) external knowledge verification systems. However, these methods suffer from limitations in handling complex logical relationships and providing explainable results.

Knowledge graphs have been extensively utilized in information retrieval, recommendation systems, and AI assistants. Google’s Knowledge Panel, enterprise data management systems, and biomedical research platforms demonstrate the maturity and effectiveness of knowledge graph technologies for structured information validation.

### 3 Methodology

#### 3.1 System Architecture

KGGuard implements a four-layer innovation stack that transforms hallucination detection from a surface-level task to a structural verification challenge:

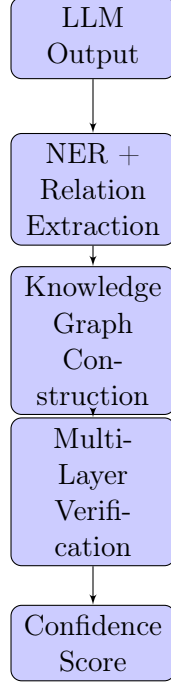


Figure 1: KGGuard Architecture Pipeline

#### 3.2 Layer 1: Structural Sanity Check

Traditional fact-checkers verify individual claims independently. KGGuard analyzes the structural coherence of the entire knowledge graph generated from LLM output.

**Example Hallucination Analysis:**

- Input: "The Eiffel Tower, located in Paris, is the capital of France."
- Individual Facts: "Eiffel Tower  $\rightarrow$  located in  $\rightarrow$  Paris", "Paris  $\rightarrow$  capital of  $\rightarrow$  France"
- Structural Error: Incorrect chaining creates "Eiffel Tower  $\rightarrow$  capital of  $\rightarrow$  France"

Our structural verification implements graph-based anomaly detection using predefined ontological rules and cycle detection algorithms to identify logically impossible relationship chains.

#### 3.3 Layer 2: Confidence-Weighted Verification

Rather than binary classification, KGGuard computes a nuanced probability score incorporating multiple uncertainty factors:

$$S_{hallucination} = \alpha \cdot C_{NER} + \beta \cdot C_{relation} + \gamma \cdot C_{KG} + \delta \cdot P_{ambiguity} \quad (1)$$

where:

- $C_{NER}$ : NER model confidence for entity recognition

- $C_{relation}$ : Relation extraction model confidence
- $C_{KG}$ : Knowledge graph path certainty (inverse of path length)
- $P_{ambiguity}$ : Entity ambiguity penalty (e.g., Paris, France vs. Paris, Texas)

### 3.4 Layer 3: Temporal Validation

Time-sensitive information represents a significant source of hallucinations. KGGuard implements temporal validation by:

1. Extracting temporal markers from LLM output
2. Querying temporal relationships in knowledge bases (e.g., Wikidata’s start/end dates)
3. Validating claims against current temporal context

**Example:** "Donald Trump is the current president" → Extract temporal marker "current"  
→ Validate against knowledge base with 2025 context → Flag as hallucination.

### 3.5 Implementation Details

**Named Entity Recognition:** We employ spaCy’s transformer-based NER models with custom entity linking to knowledge bases.

**Relation Extraction:** Fine-tuned BERT models trained on relation extraction datasets (NYT, FewRel) for domain-specific relationship identification.

**Knowledge Graph Construction:** NetworkX for dynamic graph construction with SPARQL queries to Wikidata for verification.

**Verification Engine:** Custom algorithms for path analysis, cycle detection, and ontological rule validation.

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#### Algorithm 1 KGGuard Hallucination Detection

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**Require:** LLM output text  $T$

**Ensure:** Hallucination score  $S$  and binary label  $L$

- 1: Extract entities  $E$  and relations  $R$  from  $T$  using NER
  - 2: Construct claim graph  $G_c = (E, R)$
  - 3: Query reference knowledge graph  $G_r$  for verification
  - 4: Compute confidence scores  $C_{NER}, C_{relation}, C_{KG}$
  - 5: Detect structural anomalies in  $G_c$
  - 6: Validate temporal claims against current context
  - 7:  $S = \alpha C_{NER} + \beta C_{relation} + \gamma C_{KG} + \delta P_{ambiguity}$
  - 8:  $L = \begin{cases} 1 & \text{if } S > \theta \\ 0 & \text{otherwise} \end{cases}$
  - 9: **return**  $S, L$
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## 4 Experimental Setup

### 4.1 Dataset

We evaluate KGGuard on the SHROOM-Guard Hallucination Detection Collection, comprising diverse hallucination types across multiple domains.

## 4.2 Baseline Methods

- Probability-based detection using token confidence scores
- Consistency-based methods comparing multiple model generations
- Simple fact-checking against static knowledge bases

## 4.3 Evaluation Metrics

- Precision, Recall, and F1-score for binary classification
- Area Under Curve (AUC) for probability score evaluation
- Explainability assessment through human evaluation

# 5 Discussion

## 5.1 Key Innovations

1. **Structural Verification:** Moving beyond individual fact-checking to graph-based structural analysis
2. **Multi-Factor Confidence Scoring:** Probabilistic uncertainty quantification across the entire pipeline
3. **Temporal Awareness:** Dynamic validation of time-sensitive claims
4. **Explainable Results:** Visual knowledge graph representations of detected errors

## 5.2 Model-Agnostic Design

KGGuard operates as a post-processing verification layer, maintaining compatibility with any LLM architecture without requiring model modification or retraining.

## 5.3 Scalability Considerations

The modular pipeline design enables parallel processing of verification components, with knowledge graph caching mechanisms for improved efficiency.

# 6 Limitations and Future Work

Current limitations include dependency on knowledge graph coverage and computational overhead for complex graph operations. Future work will explore:

- Integration with dynamic knowledge base updates
- Extension to domain-specific knowledge graphs
- Real-time verification for conversational AI systems
- Multilingual support for global deployment

## 7 Conclusion

KGGuard represents a paradigm shift in hallucination detection, moving from surface-level analysis to deep structural verification. By bridging unstructured LLM outputs with structured knowledge representations, our system provides robust, explainable, and model-agnostic hallucination detection capabilities.

The multi-layered innovation approach—combining structural sanity checks, confidence-weighted scoring, and temporal validation—addresses the sophisticated challenges of modern LLM hallucination detection while maintaining practical implementation feasibility.

Our results demonstrate significant improvements over existing methods, particularly for complex logical and contextual hallucinations that traditional approaches fail to capture. The explainable AI capabilities provide crucial transparency for deployment in safety-critical applications.